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### Round Robin CPU Scheduling Algorithm

Write a C program to implement the Round Robin (RR) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), Burst Times (BT), and a Time Quantum (TQ). The CPU should rotate through the processes in the ready queue, granting each a maximum of one TQ of execution time. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and the overall averages.

**Input Format**

- The first line of input contains an integer representing the number of processes.
- The second line contains an integer representing the Time Quantum.
- The next lines contain two integers for each process, where
  - The first integer denotes the Arrival Time (AT)
  - The second integer denotes the Burst Time (BT) of the process.

**Output Format**

The output displays the Average Turnaround Time and the Average

Select Language : C (GCC 9.2.0)

```
1 #include <stdio.h>
2 int main()
3 {
4     int n,t;
5     printf("Enter number of processes: \n");
6     scanf("%d",&n);
7     printf("Enter Time Quantum: \n");
8     scanf("%d",&t);
9     int AT[n],BT[n],CT[n],TAT[n],WT[n],rt[n];
10    for(int i=0;i<n;i++)
11    {
12        printf("Enter AT and BT for P%d: \n",i+1);
13        scanf("%d %d",&AT[i],&BT[i]);
14        rt[i]=BT[i];
15    }
16
17    int time=0,completed=0;
18    while(completed<n)
19    {
20        int found=0;
21        for(int i=0;i<n;i++)
22        {
23            if(AT[i]<=time && rt[i]>0)
24            {
```

Test Cases 2

Run Sample CasesRun All Cases

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Output Format

The output displays the Average Turnaround Time and the Average Waiting Time.

Sample Input 1

3  
2  
0 5  
1 3  
2 1

Sample Output 1

Enter number of processes:  
Enter Time Quantum:  
Enter AT and BT for P1:  
Enter AT and BT for P2:  
Enter AT and BT for P3:

Select Language : C (GCC 9.2.0)

```
25     found=1;
26     if(rt[i]>t)
27     {
28         time=time+t;
29         hnn
         rt[i]=rt[i]-t;
30     }
31     else
32     {
33         time=time+rt[i];
34         CT[i]=time;
35         rt[i]=0;
36         completed++;
37     }
38     }
39 }
40 if(found==0)
41 {
42     time++;
43 }
44 }
45 float totalTAT=0,totalWT=0;
46 for(int i=0;i<n;i++)
47 {
48     TAT[i]=CT[i]-AT[i];
49     WT[i]=TAT[i]-rt[i];
```

Test Cases 2

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sample input 1

3  
2  
0 5  
1 3  
2 1

Sample Output 1

Enter number of processes:  
Enter Time Quantum:  
Enter AT and BT for P1:  
Enter AT and BT for P2:  
Enter AT and BT for P3:  
  
Average Turnaround Time: 6.33  
Average Waiting Time: 3.33

Select Language : C (GCC 9.2.0)

```
36         completed++;
37     }
38 }
39 }
40 if(found==0)
41 {
42     time++;
43 }
44 }
45 float totalTAT=0,totalWT=0;
46 for(int i=0;i<n;i++)
47 {
48     TAT[i]=CT[i]-AT[i];
49     WT[i]=TAT[i]-BT[i];
50     totalTAT=totalTAT+TAT[i];
51     totalWT=totalWT+WT[i];
52 }
53 float avgTAT=totalTAT/n;
54 float avgWT=totalWT/n;
55 printf("\nAverage Turnaround Time: %.2f\n",avgTAT);
56 printf("Average Waiting Time: %.2f\n",avgWT);
57 return 0;
58 }
59 }
```

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Sample Input 1

3  
2  
0 5  
1 3  
2 1

Sample Output 1

Enter number of processes:  
Enter Time Quantum:  
Enter AT and BT for P1:  
Enter AT and BT for P2:  
Enter AT and BT for P3:  
  
Average Turnaround Time: 6.33  
Average Waiting Time: 3.33

Test Cases 2

Custom Test Case

Custom Test

Input:

3  
2  
0 5  
1 3  
2 1

Expected Output:

Enter number of processes:  
Enter Time Quantum:  
Enter AT and BT for P1:  
Enter AT and BT for P2:  
Enter AT and BT for P3:  
  
Average Turnaround Time: 6.33  
Average Waiting Time: 3.33

Output:

Enter number of processes:  
Enter Time Quantum:  
Enter AT and BT for P1:  
Enter AT and BT for P2:  
Enter AT and BT for P3:

Success

Run Sample Cases

Run All Cases

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